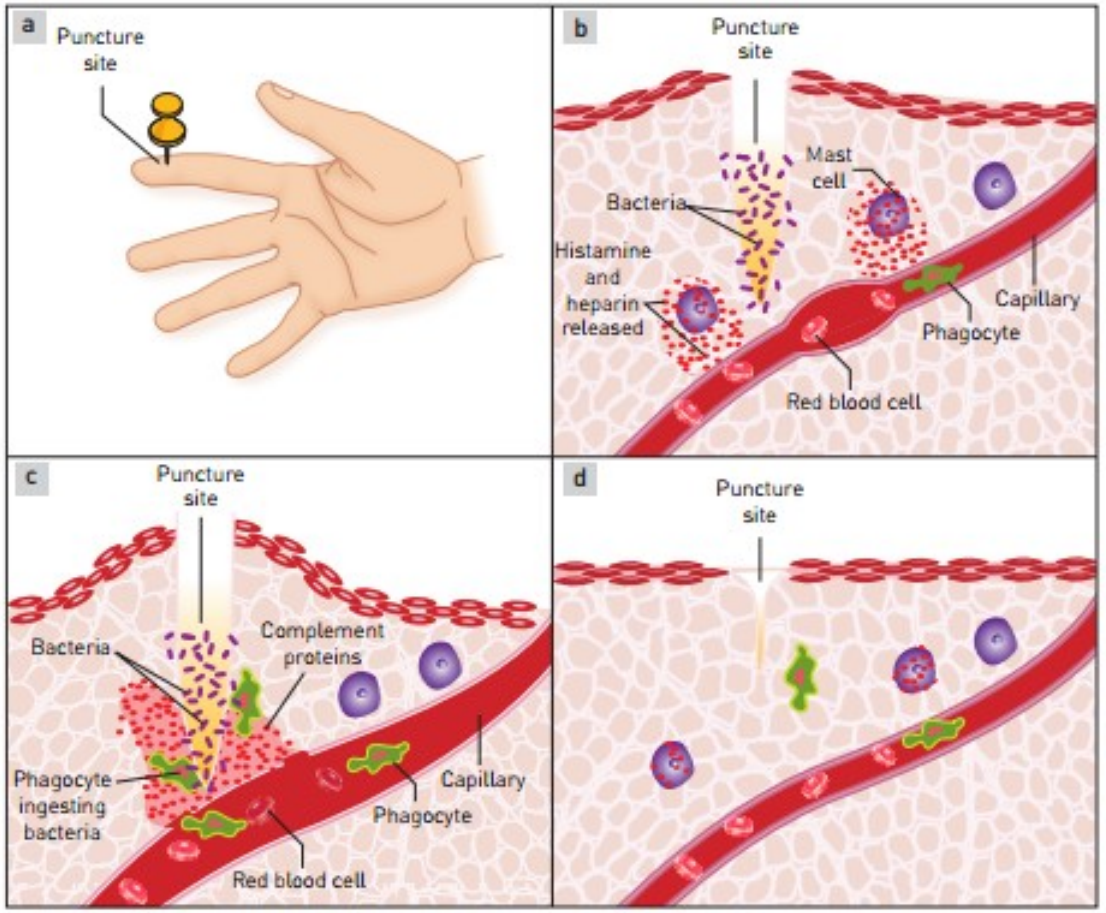
Internal Non-Specific Defences



Phagocytosis, Inflammation, Fever, Lymphatic System

Internal Non-Specific Defences

- Work against any pathogen that enters the body
- Not specific to any one pathogen
- Broad, generalised reaction to foreign substances or tissue damage.
  - Natural Killer Cells (NKC)
  - Phagocytosis
  - Inflammation
  - Fever
  - The Lymphatic System

Learning Aim: Define Non-Specific Defences

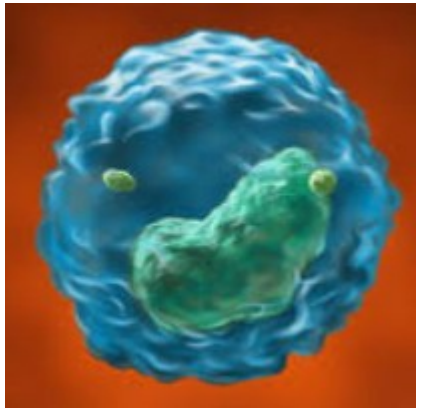
Non- Specific Defences: Natural Killer Cells (NKC)

- A type of white blood cell (leucocyte) that kills everything they encounter by releasing chemicals:
  - Cytokines – alert and attract other immune cells
  - Other chemicals that destroy cell membranes to kill unhealthy cells.

- Why don't they kill healthy body cells?
  - Healthy body cells have proteins on their surface that act as "self" markers so they don't get attacked.
  - Virus infected cells and cancer cells have fewer of these markers, so are more likely to be killed.

- Related to lymphocytes.

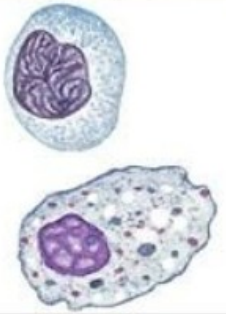
Note: there's another type of lymphocyte called a *killer T-cell* that does a similar job and is related but not identical to NKC



Learning Aim: Describe the role of Natural Killer Cells in non-specific defence against pathogens

Non-Specific Defences: Phagocytosis

- Phagocytes:
  - Cells that can **engulf and digest** micro-organisms and cell debris.
  - Leucocytes (white blood cells – many types – eg Neutrophils, Eosinophils, Monocytes)
    - In blood stream
    - Can leave blood and enter tissues at site of injury/disease
  - Macrophages
    - Some wander through tissues
    - Some are fixed

	Basophils and mast cells	*Neutrophils*	*Eosinophils*	*Monocytes and macrophages*	*Lymphocytes and plasma cells*	*Dendritic cells*
						
Primary function(s)	Release chemicals that mediate inflammation and allergic responses	Ingest and destroy invaders	Destroy invaders, particularly antibody-coated parasites	Ingest and destroy invaders Antigen presentation	Specific responses to invaders, including antibody production	Recognize pathogens and activate other immune cells by antigen presentation

Learning Aim: Describe the role of Phagocytes in non-specific defence against pathogens

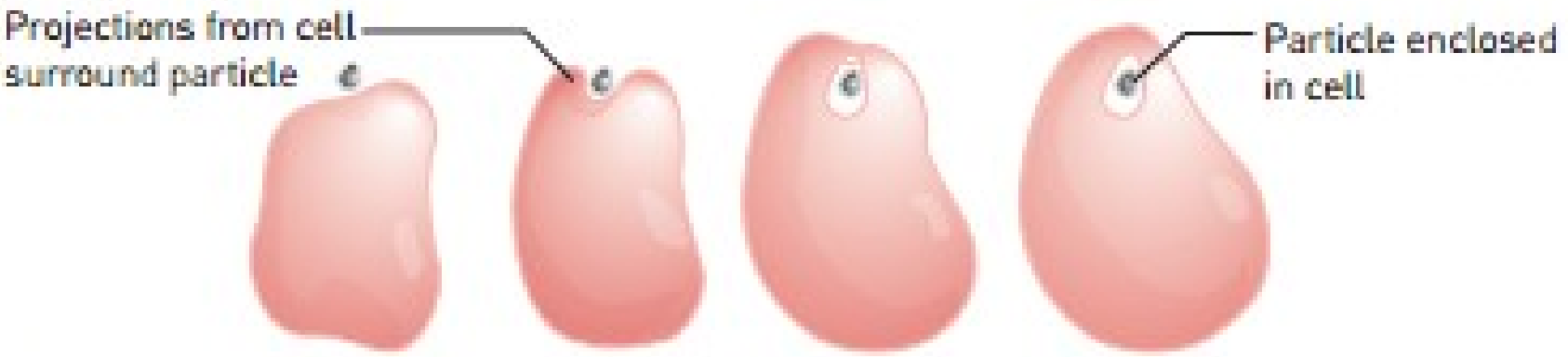
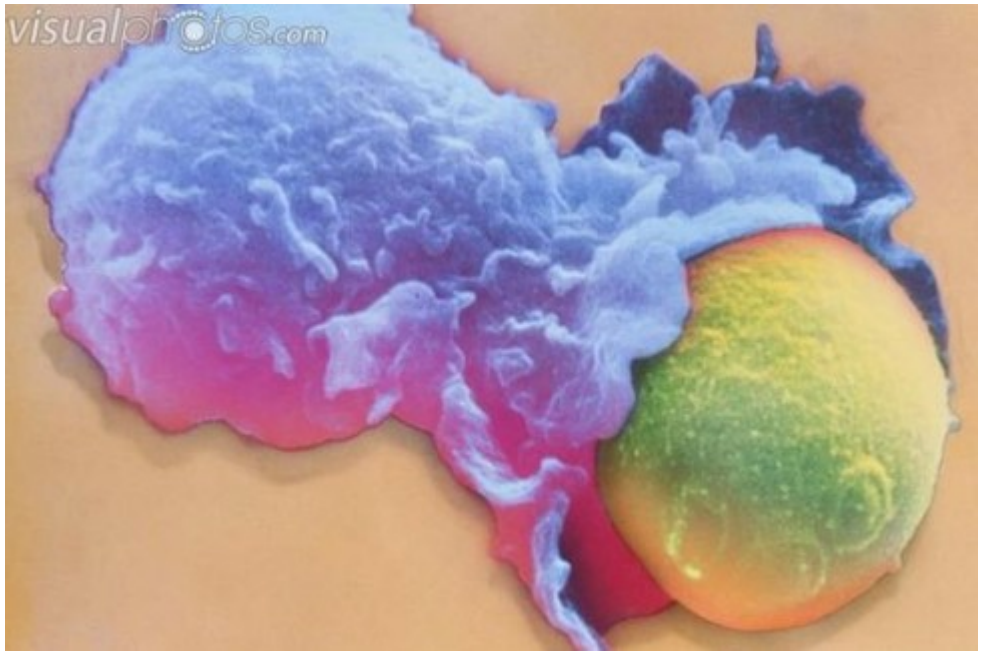


Figure 10.5 The process of phagocytosis



Non-Specific Defences: Inflammation

- Response to any tissue damage
  - Reduces spread of pathogens
  - Removes damaged tissue and cell debris
  - Begins repair of damaged tissue

- Suffix –*itis* refers to inflammation
  - Tonsillitis, laryngitis, meningitis etc

- Four signs of inflammation
  - Redness
  - Heat
  - Swelling
  - Pain



Learning Aim: Describe the role and mechanism of action of inflammation in non-specific defence against pathogens

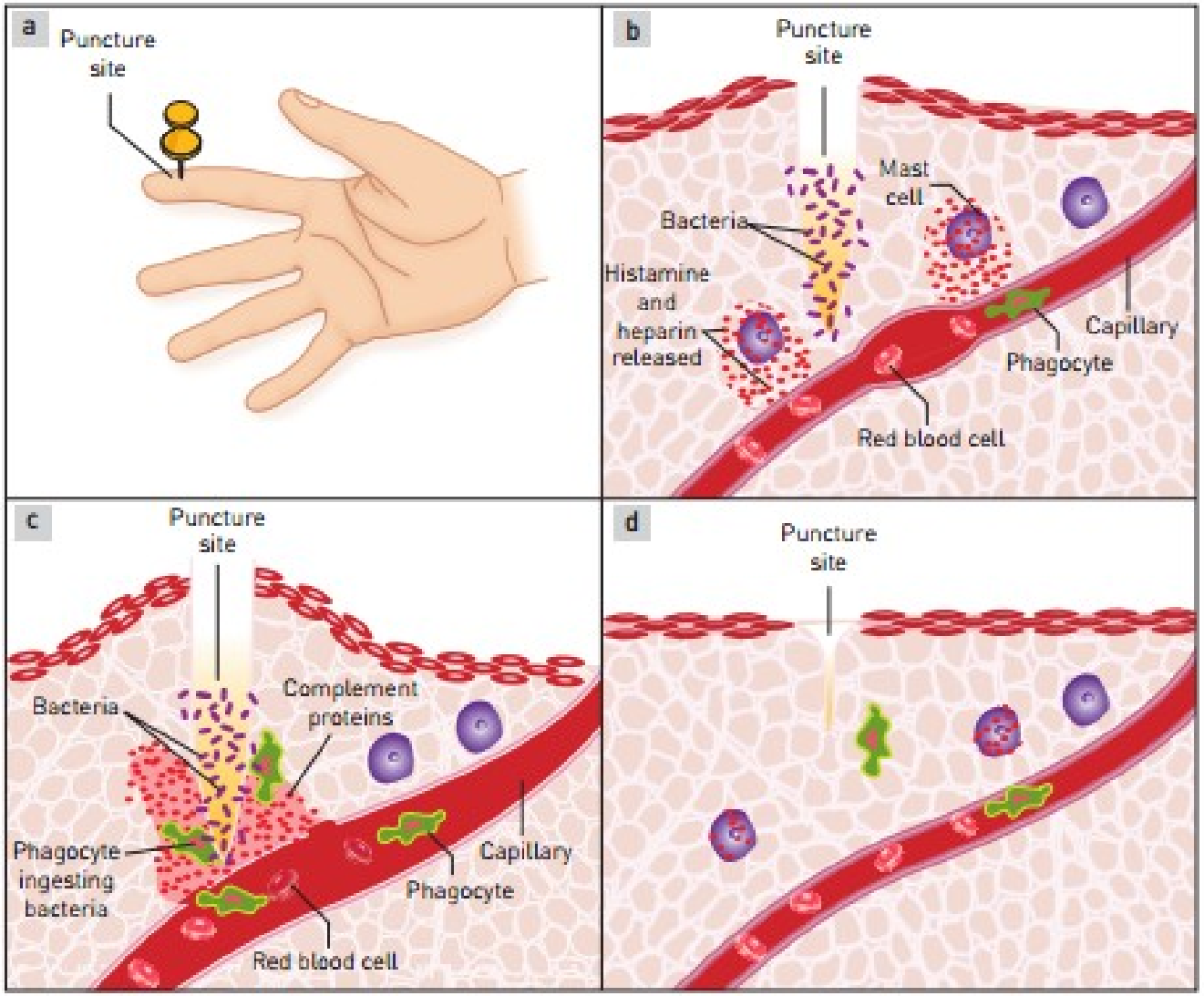


Figure 10.7 The inflammatory response: **a** When the skin is broken, a non-specific inflammatory response is triggered. **b** Mast cells release histamine and heparin. Histamine diffuses into capillaries, causing them to dilate and become leaky. The area becomes red and swells. Heparin prevents clotting in the immediate area. **c** Certain proteins are activated and attract phagocytes to the area, which engulf and digest dead cells and bacteria. **d** The tissue heals when histamine and protein signalling finish and phagocytes are no longer attracted to the area.

Steps in the Process of Inflammation

a: Tissue damage occurs.

- Pathogens may or may not be introduced to continue damage.

b: Tissue Damage causes Mast cells release histamine and heparin.

- Histamine causes local capillaries to dilate and become leaky, allowing phagocytes and other leukocytes to enter the area.
- This causes *heat*, *redness and swelling and pain*.
- Heparin prevents clotting in the immediate area.

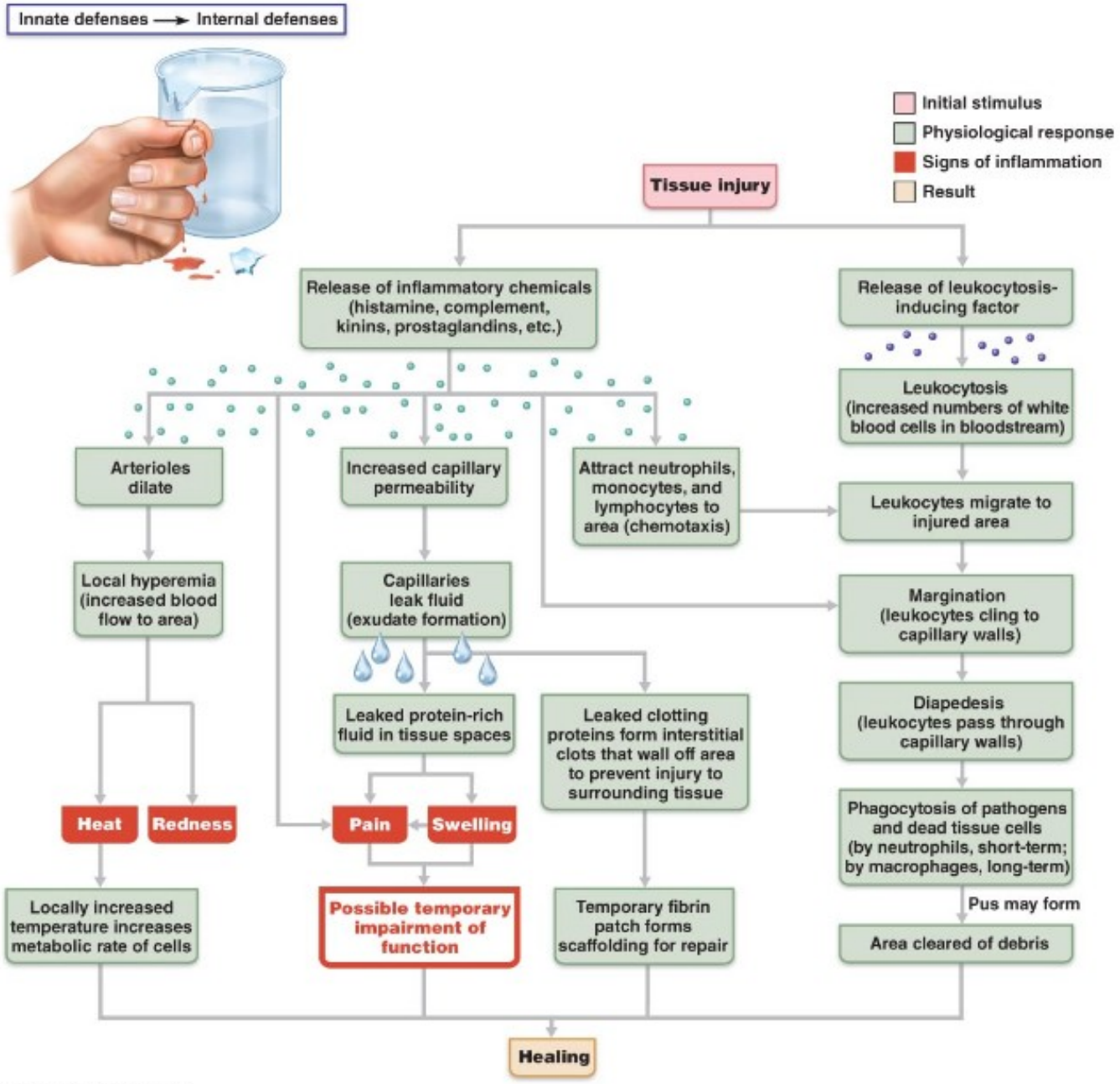
c: Complement proteins are activated, and attract phagocytes.

- They are able to enter the area of damage due to the leaky, dilated capillaries from step b.
- Phagocytes engulf and digest dead cells and bacteria.
- *Fun Fact: pus is made from dead and dying cells and bacteria!*

d: Once pathogens are cleared, mast cells and complement stop messaging

- Local capillaries return to normal
- Phagocytosis ceases
- Stage is set for tissue healing.

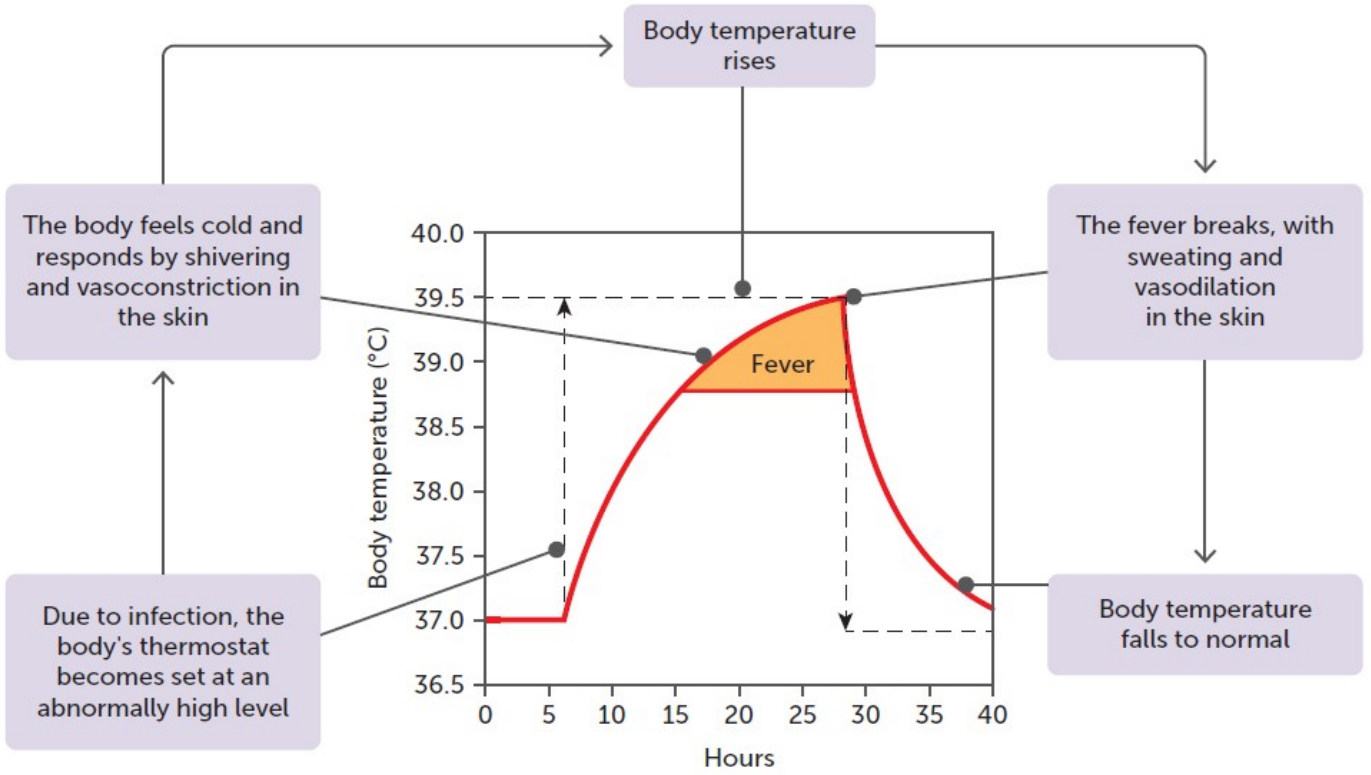
Learning Aim: Describe the role and mechanism of action of inflammation in non-specific defence against pathogens



A more complex diagram of the events of inflammation after tissue injury. You DO NOT need to know this information in this detail. It's just for additional perspective.

Non-Specific Defences: Fever (Pyrexia)

- Part of the body's natural immune response to infection.
- Body raises temperature to combat pathogens such as viruses or bacteria.
- After encountering infection, white blood cells release *pyrogens* (fever causing chemicals) eg: interleukin-1
- Pyrogens affect the hypothalamus so that the normal set point of 37°C is temporarily reset to a higher temperature.
- Hypothalamus then initiates warming responses to raise body temperature to the new, temporary set point.
- These include normal warming responses such as shivering, vasoconstriction, seeking warmth etc, until the body temperature reaches the new "set point".
- Once the infection has been dealt with, the immune system stops producing pyrogens, and the "set point" returns to normal.
- The person then feels hot and cooling responses are activated eg sweating and vasodilation.
- Fever is beneficial, but only up to a point. Prolonged, high fever can cause harm and even death, if the immune system is unable deal with the infection quickly and effectively.



Learning Aim: Discuss why fever occurs and the physiological processes involved.

Non-Specific Defences: The Lymphatic System

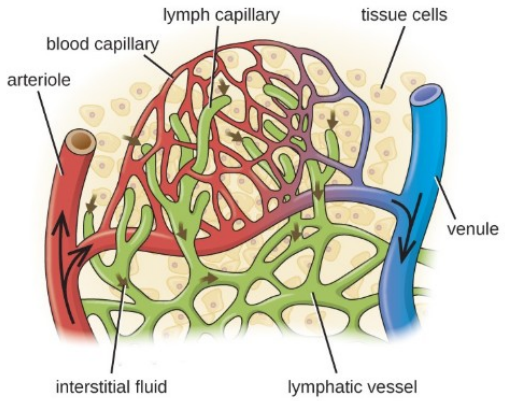
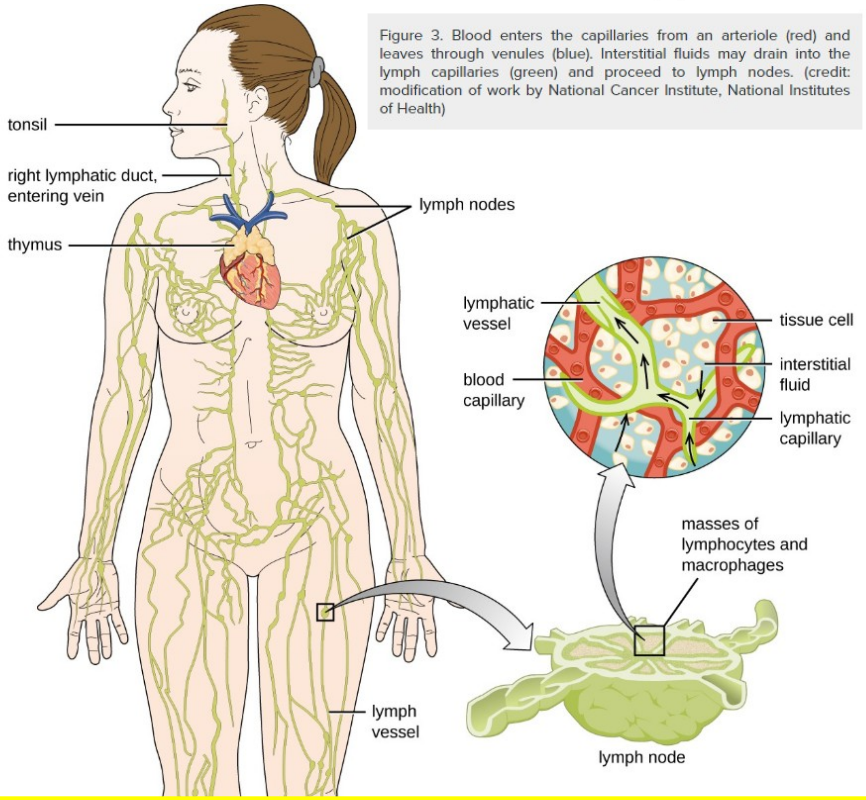
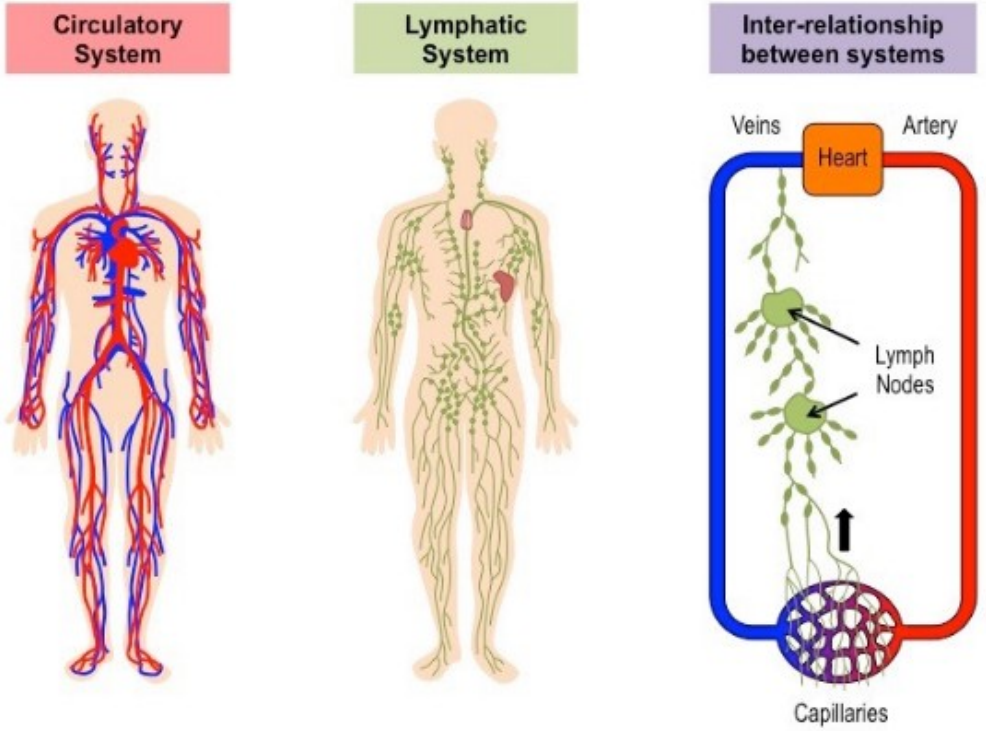


Figure 3. Blood enters the capillaries from an arteriole (red) and leaves through venules (blue). Interstitial fluids may drain into the lymph capillaries (green) and proceed to lymph nodes. (credit: modification of work by National Cancer Institute, National Institutes of Health)

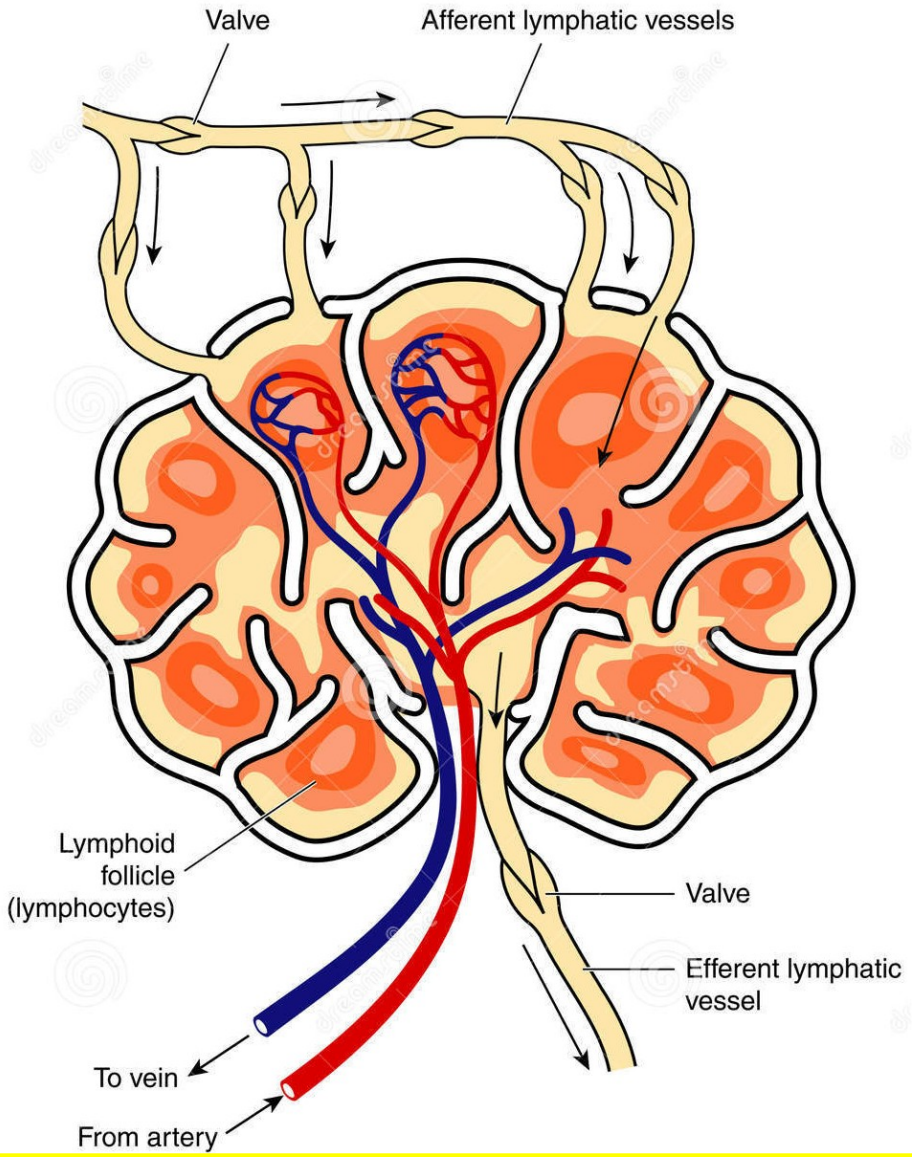


- Network of vessels
- Collects fluid in intracellular space, that has escaped capillaries
- Moves it through a series of vessels (no pump – uses squeezing of skeletal muscles during body movement)
- lymph nodes filter lymph and trap pathogens
- Returns it to blood circulation



Learning Aim: Describe the role and mechanism of action of the lymphatic system in non-specific defence against pathogens

Non-Specific Defences: The Lymphatic System – Lymph Nodes



Lymph nodes

- Filter lymph
- Process cell debris, foreign particles, microorganisms
- Contain ***lymphoid follicles*** that contain WBC – esp lymphocytes and macrophages)
- Crisscrossed by fibres to trap particles
- Can become swollen and inflamed during infection
- Lymphocytes are also important in specific immunity

Learning Aim: Describe the role of lymph nodes in non-specific defence against pathogens